

BLACK SEA TECHNOLOGY PTE. LTD.

-High-Speed Interconnection Products and Solutions

Corporate Profile

Focused on high-speed interconnection, empowering AI intelligent computing and data center, we provide highly reliable and highly compatible solutions with 1.6T/800G optical modules, 800G copper cables and active optical cables.

About Us

As a professional provider of high-speed interconnection products and solutions, we focus on the R&D, production and sales of 1.6T optical modules, 800G optical modules, 800G copper cables and active optical cables, serving core scenarios such as AI intelligent computing centers, supercomputing HPC and large cloud data centers.

Our products are fully compatible with mainstream equipment and comply with international industry standards. Relying on mature R&D, stringent quality control and efficient production, we provide reliable products for global customers and are committed to becoming a trustworthy partner in the high-speed interconnection field.

Product Center

1.6T Optical Module Series

Product Introduction

As the core product of next-generation high-speed interconnection, it adopts OSFP DR8 IHS package with a single-port bandwidth of 1700Gbps, supports PAM4 modulation, adapts to high-end scenarios such as high-density data centers and AI training clusters, and features hot pluggable and DDMI digital diagnosis, complying with environmental protection regulations.

Product Advantages

- High bandwidth: 1600Gbps transmission rate, meeting the demand for massive data exchange and breaking the bandwidth bottleneck.
- Stable and low power consumption: Power consumption $\leq 25W$, adapting to the high-temperature and high-density environment of Data Centers with reliable long-term operation.
- Wide compatibility: Compliant with international standards, compatible with mainstream equipment, plug-and-play for convenient operation and maintenance.

Technical Parameters

Parameter	Specification
Transmission Rate	1600Gbps, 8×106.25GBd PAM4
Form factor	OSFP DR8 IHS (Hot Pluggable)
Power Consumption/Temperature	$\leq 25W$, 0~70°C
Transmission Distance	500m (G.652 single-mode fiber)
Core Functions	DDMI digital diagnosis, hot pluggable

Application Fields

AI training/inference clusters, supercomputing HPC, high-density data centers, 1600G Ethernet and other scenarios.

800G Optical Module Series

Product Introduction

As the main transmission product for AI intelligent computing and data centers, it covers two models: OSFP DR4 and OSFP DR8. Adopting PAM4 modulation, it supports hot pluggable and DDMI diagnosis, adapts to short and medium-distance transmission, and helps customers reduce TCO.

Product Advantages

- High density and stability: Nominal transmission rate of 850Gbps, adapting to high-density deployment in data centers with stable signal transmission.
- Flexible adaptation: Two models optional, power consumption $\leq 18W$, supporting multi-mode applications.
- Compatible and reliable: Compliant with international standards, compatible with mainstream equipment, and quality tested stringently.

Technical Parameters

Parameter	OSFP DR4	OSFP DR8
Transmission Rate	850Gbps, 4×106.25GBd PAM4	850Gbps, 8×53.125GBd PAM4
Power Consumption	$\leq 18W$	$\leq 16W$
Transmission Distance	500m (G.652 single-mode fiber)	500m (G.652 single-mode fiber)
Core Functions	DDMI diagnosis, hot pluggable	DDMI diagnosis, hot pluggable

Application Fields

AI intelligent computing centers, supercomputing HPC, large cloud data centers, adapting to short and medium-distance transmission needs such as inter-switch connection and GPU cluster interconnection.

800G Copper Cable and Active Optical Cable Series

Product Introduction

Including active AEC/ACC, passive DAC and AOC types with OSFP Form factor, it is a cost-effective choice for short-distance interconnection in data centers, featuring convenient deployment and stable transmission.

Product Advantages

- High cost performance: Cost-effective, plug-and-play, simplifying wiring and reducing deployment costs.
- Stable transmission: Active series integrate signal conditioning chips, while passive series have excellent shielding performance with low signal attenuation.
- Flexible wiring: Multiple lengths optional, adapting to high-density cabinet layout, compliant and reliable.

Technical Parameters

Cable Type	Transmission Rate	Length Power	Consumption(Per End)
Active AEC	800Gbps	1~5m	12w
Active ACC	800Gbps	0.5~2.5m	1.2w
Passive DAC	800Gbps	0.5~1.5m	0.005w
Active Optical Cable (AOC)	800Gbps	1~50m	12w

Application Fields

Short-distance connection within data center racks and between adjacent cabinets, adapting to scenarios such as GPU and switch interconnection and structured wiring in cabinets.

Quality Assurance

Quality Philosophy

Adhering to the concept of "full life cycle quality management" and abiding by the standards of high reliability and high compatibility, we provide customers with high-quality products that meet international standards.

Quality Control

- R&D control: Following international standards, avoiding technical risks in advance to ensure product compliance from the source.
- Supply chain control: Strictly screening suppliers to ensure the consistency and reliability of raw materials.
- Production and testing control: Complying with the ISO 9001:2015 system, conducting full-process quality inspection, and delivering finished products only after passing comprehensive tests.

Certification Qualifications

Certified by the ISO 9001:2015 system, our products comply with international standards such as IEEE and CMIS, with complete test reports and in line with RoHS 2.0 environmental protection requirements.

Production and Manufacturing

Production Philosophy

With the philosophy of "Efficiency, Intelligence and Compliance", we build an intelligent manufacturing system to ensure on-time and on-quality product delivery.

Production Strength

We have a modern production base and intelligent production lines covering the entire production process; equipped with a professional team, we implement strict standardized operations to ensure production capacity and product quality.

Production Advantages

- High-efficiency production capacity: Intelligent production to meet the needs of large-scale orders.
- Controllable quality: Full-process quality inspection with strong product consistency.
- Flexible adaptation: Supporting customized production to meet diversified demands.

Solutions

800G RoCE Lossless Network Solution

Targeting the needs of AI intelligent computing and supercomputing, relying on the high bandwidth and low latency advantages of 800G products, combined with RoCEv2 technology, it improves distributed training performance and adapts to the training needs of large AI models.

Data Center High-Density Interconnection Solution

Combining 1.6T, 800G optical modules, copper cables and active optical cable products, it adapts to the leaf-spine network architecture, simplifies wiring, reduces costs and supports the expansion and upgrade of data centers.

Customized High-Speed Interconnection Solution

Providing solution design, customizing products according to customer scenarios, optimizing network architecture, reducing TCO and meeting personalized needs.

Service Support

- Warranty service: All products are provided with a 2-year warranty, and free repair and replacement are available for non-human damage.
- Technical support: 7×24 professional technical support for timely response to consultations and problem troubleshooting.
- Solution consulting: Providing solution consulting to assist in optimizing network architecture.
- After-sales tracking: Regularly tracking product usage and continuously optimizing products and services.